

Course Code : ECT 351

KQLR/RS – 24 / 1604

**Fifth Semester B. Tech. (Electronics and Communication Engineering)
Examination**

ELECTROMAGNETIC WAVES

Time : 3 Hours]

[Max. Marks : 60

Instructions to Candidates :—

- (1) All questions are compulsory.
 - (2) All questions carry marks as indicated.
 - (3) Illustrate your answer with neat sketches.
 - (4) Assume suitable data wherever necessary.

1.
 - (a) Derive the expression for impedance transformation on a transmission lines moving towards generator. 5(CO1)
 - (b) An impedance of $60 + j40 \Omega$ is connected to 50Ω line. Find the position and length of single stub impedance matching network. 5(CO1)

2.
 - (a) Derive relationship between E and H fields for uniform plane waves. 5(CO2)
 - (b) A material has dielectric constant 25 and conductivity 2×10^6 mho/m. What is the frequency above which the material cannot behave like a good conductor ? If a Plane Wave of 10 MHz is incident on a material, effectively up to what depth can the wave penetrate the material and what will be the wavelength of the wave inside the material ? 5(CO2)

3.
 - (a) Explain the Reflection and Refraction of plain waves at dielectric interface with perpendicular polarization. 5(CO4)
 - (b) A light beam is incident from air to a medium with a dielectric constant 4 and relative permeability 100. If the angle of incidence is 60° , find the angle of reflection and angle of refraction. 5(CO4)

KQLR/RS-24 / 1604

Contd.

4. (a) Prove that attenuation factor in parallel plane wave guide is $\alpha = \text{Power lost per unit length} / 2 \times \text{Power Transmitted}$. 5(CO5)
- (b) Discuss Phase velocity, Group velocity and Dispersion in Parallel Plane Waveguide. 5(CO5)
5. (a) Discuss Cut-off frequency, Cut-off wavelength in TE and TM modes of Rectangular waveguides. 5(CO5)
- (b) A rectangular wave guide of cross sectional dimension $2.3 \text{ cm} \times 1 \text{ cm}$ is used to transmit a signal of 9 GHz. Determine :
- (i) Cutoff frequency for the dominant mode.
 - (ii) Phase and group velocities of dominant mode.
 - (iii) Guide wavelength. 5(CO5)
6. (a) Explain the following terms related to antenna :
- (i) Radiation Pattern.
 - (ii) Null Power Beam Width.
 - (iii) Half Power Beam Width.
 - (iv) Directivity.
 - (v) Gain. 5(CO3)
- (b) Estimate the received power if the transmitted power is 100 W. Transmitting and Receiving antenna operating at 3 GHz with Gains of 30 dB and 20 dB respectively. Both are separated by a Distance of 1 km. Assume that both antennas are identical and have same Polarization. 5(CO3)

